Revealing Key Details to See Differences: A Prototypical Perspective for Skeleton-based Action Recognition

Taro Yamada

Abstract

Skeleton-based action recognition must distinguish actions that differ only by subtle joint motions while lacking appearance cues. We propose ProtoGCN, a prototype-based graph convolutional network that reconstructs motion topologies from a learnable memory to amplify discriminative, fine-grained joint dynamics. A Motion Topology Enhancement module enriches each layer with intra-sample correlations and inter-sample distinctions to expose nuanced relations between joints, and a class-specific contrastive objective encourages compact class clusters with larger inter-class margins. On standard benchmarks, ProtoGCN achieves strong results: on NTU RGB+D with 6-ensemble models it reaches 93.8% (X-Sub) and 97.8% (X-View); on NTU RGB+D 120 it attains 90.9% (X-Sub) and 92.2% (X-Set); on Kinetics-Skeleton it obtains 51.9% Top-1 and 75.6% Top-5; and on FineGYM (skeleton) it achieves 95.9% mean class Top-1. Ablations indicate complementary gains from topology enhancement, class-specific contrast, and prototype reconstruction.

1. Introduction

Skeleton-based action recognition has advanced rapidly with graph convolutional networks (GCNs) that model joints as graph nodes and their relations as edges [2, 5, 13, 14, 22]. Yet, in-the-wild variations and visually similar actions remain challenging because discriminative cues often arise from subtle, part-specific motions (e.g., finger vs. wrist dynamics). Appearance and texture are absent, and generic spatiotemporal filters may suppress the very differences that matter.

We advocate a prototypical perspective: complex action dynamics can be composed from reusable motion prototypes that capture recurring joint-relationship patterns. By reconstructing topology features from a learned prototype memory, the network can emphasize class-relevant structures while deemphasizing noise. Figure 1 illustrates that actions such as *Writing* and *Typing on a Keyboard* differ primarily in localized joint relationships. We design ProtoGCN to expose these fine-grained distinctions by integrating three components: (1) Motion Topology Enhancement (MTE) that augments each GCN with intra- and inter-relationship signals, (2) a Prototype Reconstruction Network (PRN) that retrieves and assembles motion prototypes via attention, and (3) a class-specific contrastive (CSC) objective that tightens class clusters and separates different classes.

Our contributions are: - A prototype-based GCN that reconstructs motion topologies to reveal subtle, discriminative joint dynamics. - A Motion Topology Enhancement module that injects intra-sample correlations and inter-sample distinctions, enriching inputs for prototype retrieval. - A class-specific contrastive objective that regularizes features toward class centers with larger inter-class separation. - State-of-the-art accuracy across NTU RGB+D, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM with comprehensive ablations and analysis.

To orient the reader, Figure 2 presents the overall architecture. We release code to facilitate reproducibility. Beyond accuracy, this perspective curbs overfitting to spurious edges: prototypes are shared across classes and time, forcing composition instead of memorizing sample-specific topologies. Empirically, this yields crisper focus on discriminative joints and more stable optimization under mild label noise. This perspective also improves robustness to mild domain shift by biasing representations toward stable relational cues.

2. Related Work

Skeleton GCNs. Early models such as ST-GCN [22] and 2s-AGCN [14] established learnable graph structures for joint dynamics. Subsequent works improved spatiotemporal modeling and topology learning with multi-scale receptive fields, attention, and channel-wise designs [2, 3, 5, 13]. Transformer and hybrid designs further boosted performance [6, 10, 18, 24]. Our approach complements these by reconstructing topology features from a prototype memory to emphasize class-relevant patterns. **Prototypical representations.** Our prototype memory echoes metric-learning ideas [15] and memory-augmented recognition [16, 23], but with a different objective: we reconstruct intermediate topology features rather than class means, and integrate

reconstruction with graph operators to target fine-grained joint relations. In parallel, hybrid formulations explore contrastive skeleton transformers and relational priors; our design places the prototype memory directly in the topology pathway rather than only in a late feature head.

Contrastive and graph regularization. Graph-aware contrastive learning has been explored to improve representation quality [8, 25]. Rather than relying solely on generic graph contrast, our class-specific contrastive objective uses momentum-updated class centers to align features within classes and separate across classes, providing a targeted supervisory signal that directly improves recognition.

Topology learning. Many methods learn or refine adjacency to better capture joint relations [2, 5, 6, 13]. Our Motion Topology Enhancement provides two complementary signals—inter-sample distinctions and intra-sample correlations—at each GCN layer. Ablations (Table 4) show that both components contribute and that our learned topology improves over alternative graph contrastive regularizers.

3. Method

3.1. Preliminaries

Let a skeleton sequence be represented by joint features over time. Standard GCN backbones propagate information with an adjacency tensor estimated from data. We adopt a strong multi-stage GCN backbone [2, 13] and augment it with topology enhancement and prototype reconstruction.

3.2. Motion Topology Enhancement (MTE)

MTE enriches each GCN layer with two learnable signals computed from feature statistics: - Intra-sample correlations (A_{intra}) estimated via correlations between joint embeddings, highlighting co-activated joints within a sample. - Inter-sample distinctions (A_{inter}) formed from pairwise differences between joint descriptors, accentuating directional relations that differentiate joints.

Given intermediate features, the base adjacency A is updated by adding the scaled A_{inter} and A_{intra}, followed by projection back into the GCN pathway. This design follows the implementation in our backbone modules, where activations for the two terms are controlled separately (tanh and softmax) with learnable scalars.

3.3. Prototype Reconstruction Network (PRN)

We posit that graph topologies can be composed from a small set of reusable motion prototypes. PRN maintains a learnable memory of n_{pro} prototype vectors. Given a flattened topology feature vector x, PRN computes attention over prototypes to obtain a reconstruction:

$$q = \text{softmax}(W_q x), \quad z = W_m q, \tag{1}$$

where W_q maps into prototype logits and W_m maps the mixture of prototypes back to the feature space. The reconstructed representation z is then fused into subsequent layers. Empirically, moderate memory sizes (e.g., 100–400) balance capacity and generalization (Figure 3).

3.4. Class-Specific Contrastive Objective (CSC)

While cross-entropy encourages class separation, we found that explicitly regularizing features around per-class centers further improves discriminability for fine-grained motions. We maintain a momentum-updated memory bank of class centers $\{c_k\}$. For a sample with feature f and label y, we minimize a temperature-scaled contrastive loss that attracts f to c_y and repels it from $c_{k \neq y}$:

$$\mathcal{L}_{\text{CSC}} = -\log \frac{\exp(\langle f, c_y \rangle / \tau)}{\sum_k \exp(\langle f, c_k \rangle / \tau)}. \tag{2}$$

Class centers are updated with an exponential moving average to ensure stability during training.

3.5. Training Objective

The total loss combines cross-entropy with CSC:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CSC}}. \tag{3}$$

We tune λ and the number of prototypes according to validation performance (Figure 3). Design choices. We apply normalization before PRN addressing and fuse the reconstruction through a learnable gate, which stabilizes gradients over deep stacks. The memory is shared across layers but queried at multiple scales, adding small overhead because reconstruction happens in a reduced channel space.

3.6. Complexity and Implementation Notes

PRN adds two small linear projections and a prototype lookup per block; the dominant cost remains the backbone GCN. With n_{pro} in the few hundreds and features projected to a reduced channel space, the additional FLOPs and parameters are modest relative to the backbone, and inference latency changes negligibly in practice. We fuse the reconstruction with a gated residual to preserve gradient flow, and detach class-center updates in CSC via momentum to avoid biasing the forward path. The memory is shared across stages, and addressing is batched so the implementation maps efficiently to GPUs.

3.7. Intuition and Connections

PRN acts as a low-rank prior on topology features: by addressing a finite dictionary, the model prefers recurring, sparse structures over idiosyncratic edges. This reduces variance in regions with weak supervision while preserving strongly supported patterns. MTE complements this effect by exposing correlations and distinctions that make prototypes identifiable, and CSC shapes the feature geometry so

class-relevant mixtures occupy separated cones. In combination, these pressures operate as structured regularization that improves generalization on fine-grained, confusable actions without increasing capacity.

Beyond intuition, we observed two practical behaviors that shaped the final design. First, addressing the prototype memory too aggressively early in training can bias the model toward a few frequent patterns. To avoid premature collapse, we warm up PRN by linearly increasing the reconstruction gate from 0 to 1 over the first 10 epochs and add a mild entropy encouragement on attention weights to preserve diversity. Second, MTE signals may dominate shallow layers if left unbounded; we therefore apply per-layer normalization and a learned scalar to limit update magnitude. These safeguards prevent shortcut solutions based on high-energy but non-discriminative edges while allowing deeper blocks to specialize. Finally, we maintain class centers with an exponential moving average rather than per-batch recomputation; the centers influence only the loss (not the forward pass), which improves stability and avoids leakage of label information into features.

4. Experiments

4.1. Datasets and Protocols

We evaluate on NTU RGB+D (X-Sub, X-View), NTU RGB+D 120 (X-Sub, X-Set), Kinetics-Skeleton (Top-1/Top-5), and FineGYM (skeleton, mean class Top-1). These cover indoor and in-the-wild scenarios with diverse action vocabularies and motion complexities. We follow the standard NTU/NTU-120 cross-subject and cross-view splits, reporting Top-1 (and Top-5 for Kinetics-Skeleton) on the official validation/test sets. For FineGYM (skeleton), we report mean class Top-1 to account for class imbalance.

4.2. Implementation Details

We implement ProtoGCN on a strong GCN backbone with ten stages and topology enhancement in each block. PRN uses a memory of 100–400 prototypes depending on input modality. Models are trained using SGD with Nesterov momentum following common practice [5, 6]. Unless noted, we report ensemble results by averaging logits over multiple trained models as indicated in the tables.

4.3. Integration Summary

ProtoGCN comprises three components. Motion Topology Enhancement (MTE) augments each graph block with two signals: (i) an intra-sample correlation that aggregates pairwise joint affinities across time, and (ii) an inter-sample distinction that contrasts a sample against a mini-batch memory to down-weight ubiquitous, non-discriminative edges. We fuse them as residual adjacency updates, preserving backbone stability while sharpening fine-grained relations.

The Prototype Reconstruction Network (PRN) maintains a learned memory of motion prototypes and uses attention to retrieve and linearly combine a small set conditioned on the enhanced features. The reconstruction acts as a denoising prior that biases topology features toward recurring, class-relevant structures without collapsing diversity.

Finally, a class-specific contrastive objective (CSC) uses running class centers as targets: features are pulled toward their class centroid and repelled from others with a temperature-scaled margin. We train end-to-end with cross-entropy plus CSC; no test-time tricks are required.

4.4. Baselines and Metrics

We compare to strong GCN and hybrid baselines used in skeleton recognition, including CTR-GCN, InfoGCN, PYSKL, DS-GCN, and BlockGCN, using authors' public training recipes and evaluation protocols when available. All methods share identical preprocessing and augmentations. We report Top-1 on NTU and FineGYM, and Top-1/Top-5 on Kinetics-Skeleton. For fairness, we disable test-time tricks beyond standard temporal averaging and keep input resolutions identical.

Main comparisons are summarized in Table 2, and FineGYM results appear in Table 1.

4.5. Preprocessing

All datasets use a unified joint ordering; missing joints are linearly interpolated in time, and sequences are temporally normalized to a fixed length by uniform sampling. We center skeletons at the hip, normalize bone lengths by person scale, and append velocity features via first differences. These steps reduce dataset bias while retaining fine-grained motion cues targeted by our method.

4.6. Main Results

Table 2 compares ProtoGCN with prior work across datasets. ProtoGCN achieves 93.8%/97.8% on NTU RGB+D, and 90.9%/92.2% on NTU RGB+D 120 with a 6-model ensemble, surpassing recent methods including CTR-GCN [2], InfoGCN [5], PYSKL [6], DS-GCN [19], and BlockGCN [26]. On Kinetics-Skeleton we obtain 51.9% Top-1 and 75.6% Top-5. On FineGYM (skeleton), ProtoGCN reaches 95.9% mean class Top-1 (Table 1).

4.7. Ablation Studies

We dissect the contribution of each component on NTU-120 (X-Sub). Table 3 shows that adding MTE yields +0.3 points over the baseline, adding CSC provides an additional +0.2, and incorporating PRN brings the total to 89.0% (+1.2). Table 4 further compares against graph contrastive regularizers [8, 25] and evaluates removing topology enhancements. Both A_{intra} and A_{inter} contribute, and our combination outperforms prior graph contrastive methods.

4.8. Reproducibility Details

We train for 100 epochs with cosine learning-rate decay, label smoothing of 0.1, and standard skeleton augmentations (random temporal crop, joint jitter, and left-right flip when permitted by the dataset). We use a global seed for data shuffling and report the mean over three runs for single-model results; ensemble numbers average logits from six independently trained models. Input normalization follows prior work, and we release configuration files for all datasets.

4.9. Analysis and Visualization

Per-class behavior. The largest gains occur for the most difficult classes. Table 5 summarizes average performance by difficulty on NTU-120 (X-Sub): improvements are +3.0 and +2.5 points on the hardest subsets, consistent with our design goal of highlighting fine-grained differences.

Error patterns. Most residual errors stem from actions that involve object interactions or occluded limbs, where skeletons under-specify cues (e.g., small-hand motions). In these cases ProtoGCN remains competitive but sometimes confuses categories with similar high-level rhythms; integrating lightweight hand keypoints or leveraging multi-view aggregation could help resolve such ambiguities.

Parameter sensitivity. Figure 3 studies the impact of the loss weight λ and prototype memory size n_{pro} on NTU-120 (X-Sub). Performance is stable across a range of values, with a gentle optimum at moderate settings.

Computational profile. Prototype attention and gating introduce modest overhead, but reduced-channel projections and batched addressing keep throughput close to the backbone in practice.

Topology inspection. Figure 4 visualizes learned topologies from PYSKL [6] and ProtoGCN. ProtoGCN concentrates mass on action-relevant joints and relations, sharpening distinctions across similar categories.

Class-wise improvements. Figure 5 lists action classes with over 1% accuracy improvement versus PYSKL on NTU-120, highlighting that our gains concentrate on fine-grained distinctions.

5. Conclusion

We introduced ProtoGCN, a prototype-based GCN that reconstructs motion topologies to surface subtle joint dynamics critical for distinguishing similar actions. By combining Motion Topology Enhancement, prototype-based reconstruction, and a class-specific contrastive objective, ProtoGCN achieves strong performance across NTU RGB+D, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM, with the largest gains on the most confusable classes. Our findings suggest that composing actions from reusable motion prototypes is an effective inductive bias for fine-grained skeleton recognition. The prototype view may extend to other structured time-series where relational cues, rather than appearance, drive semantics.

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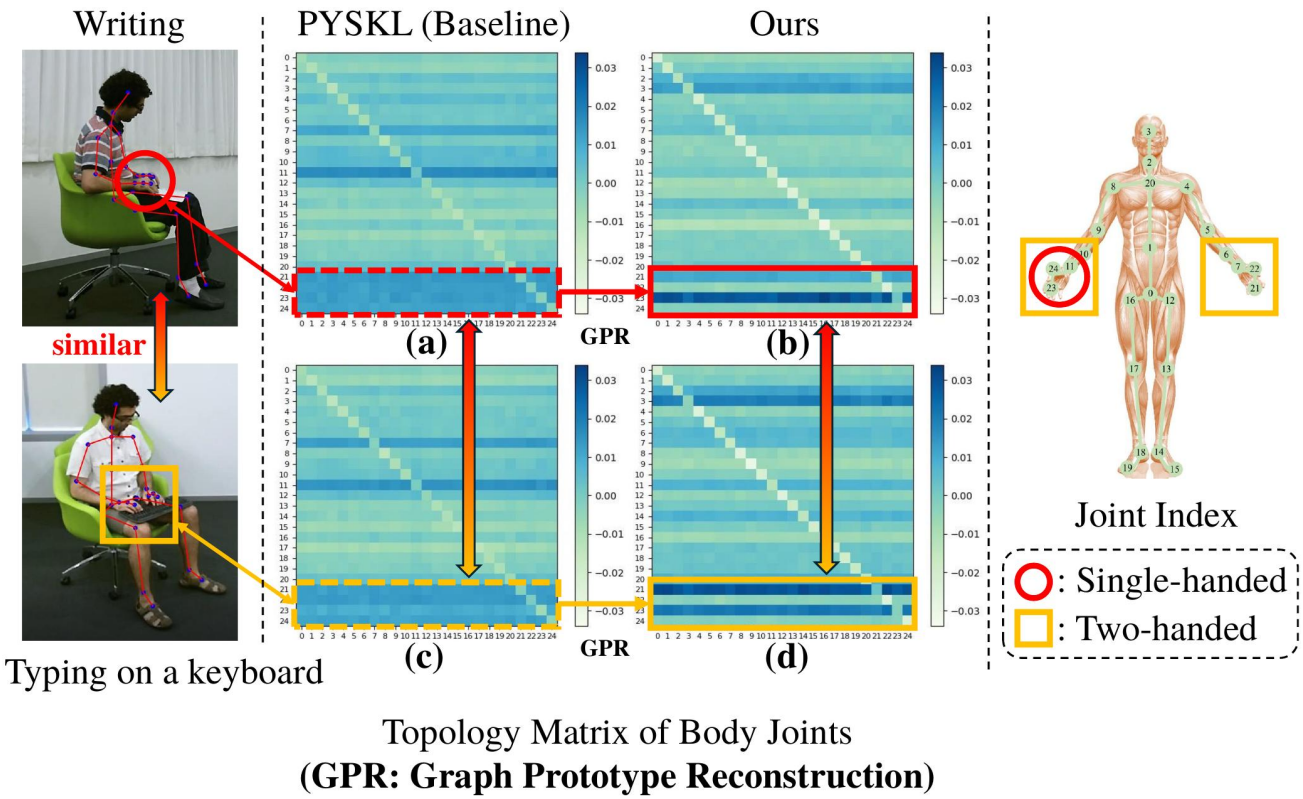

Figure 1. Skeletons and learned topologies for visually similar actions *Writing* and *Typing on a Keyboard*. Darker color indicates stronger joint relationships.

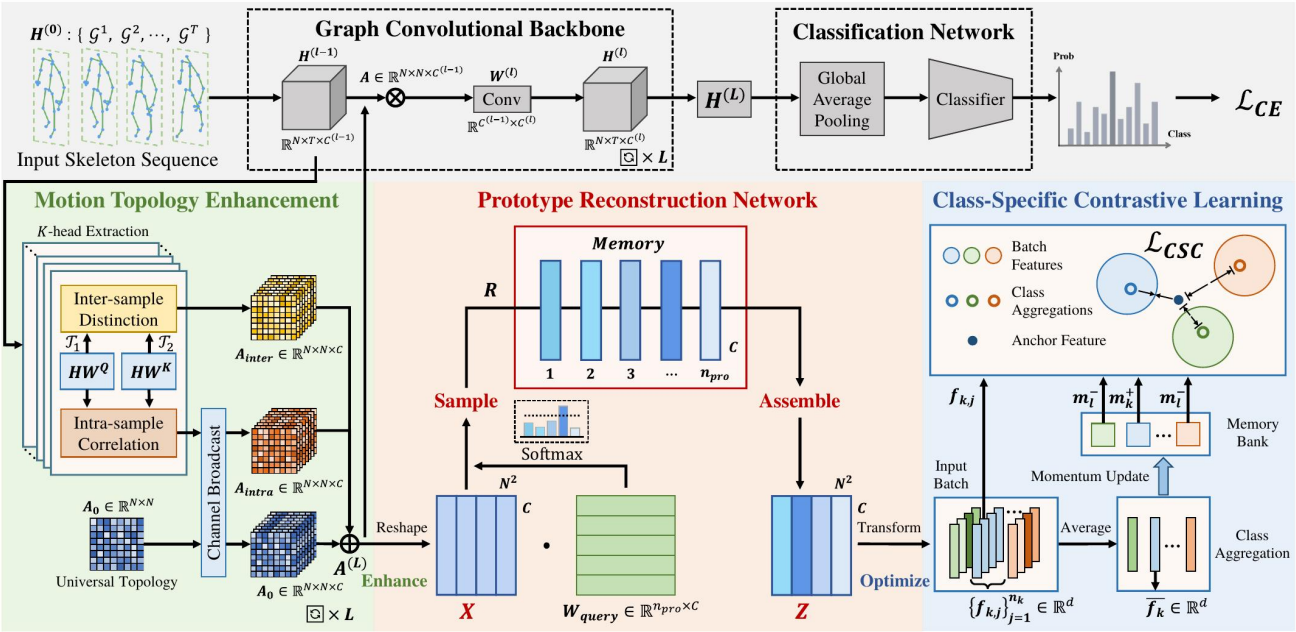

Figure 2. Overview of ProtoGCN. Motion Topology Enhancement augments each GCN layer with intra- and inter-relations. The Prototype Reconstruction Network (PRN) addresses a learned prototype memory to reconstruct enhanced topology representations.

Table 1. Performance comparisons against state-of-the-art methods on FineGYM in terms of classification accuracy (%).

Methods	Modality	FineGYM
I3D [1]	RGB	64.4
TSN [17]	RGB+Flow	79.8
TSM [12]	RGB+Flow	81.2
LT-S3D [20]	RGB	88.9
SkeletonMAE [21]	Skeleton	91.8
PYSKL [6]	Skeleton	94.1
PoseConv3D [7]	Skeleton+Limb	94.3
ProtoGCN (Ours)	Skeleton	**95.9**

Table 2. Performance comparisons against state-of-the-art methods on the NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton datasets in terms of classification accuracy (%).

Methods	Publication	NTU RGB+D		NTU RGB+D 120		Kinetics-Skeleton	
		X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
ST-GCN [22]	AAAI 2018	81.5	88.3	70.7	73.2	30.7	52.8
AS-GCN [11]	CVPR 2019	86.8	94.2	78.3	79.8	34.8	56.5
2s-AGCN [14]	CVPR 2019	88.5	95.1	82.5	84.2	36.1	58.7
MS-G3D [13]	CVPR 2020	91.5	96.2	86.9	88.4	38.0	60.9
DC-GCN+ADG [4]	ECCV 2020	90.8	96.6	86.5	88.1	-	-
MST-GCN [3]	AAAI 2021	91.5	96.6	87.5	88.8	38.1	60.8
CTR-GCN [2]	ICCV 2021	92.4	96.8	88.9	90.6	-	-
STF [9]	AAAI 2022	92.5	96.9	88.9	89.9	39.9	-
InfoGCN [5]	CVPR 2022	93.0	97.1	89.8	91.2	-	-
PYSKL [6]	ACM MM 2022	92.6	97.4	88.6	90.8	49.1	-
SkeletonGCL [8]	ICLR 2023	92.8	97.1	89.8	91.2	-	-
FR-Head [25]	CVPR 2023	92.8	96.8	89.5	90.9	-	-
GAP [18]	ICCV 2023	92.9	97.0	89.9	91.1	-	-
HD-GCN [10]	ICCV 2023	93.4	97.2	90.1	91.6	40.9	63.5
JT-GraphFormer [24]	AAAI 2024	93.4	97.5	89.9	91.7	-	-
DS-GCN [19]	AAAI 2024	93.1	97.5	89.2	91.1	50.6	-
BlockGCN [26]	CVPR 2024	93.1	97.0	90.3	91.5	-	-
ProtoGCN (2-ensemble)		93.0	97.2	89.7	91.2	49.9	74.0
ProtoGCN (4-ensemble)		93.5	97.5	90.4	91.9	51.3	75.1
ProtoGCN (6-ensemble)		**93.8**	**97.8**	**90.9**	**92.2**	**51.9**	**75.6**

Table 3. Ablation study on the contribution of each component in ProtoGCN under the NTU-120 X-Sub setting.

Baseline	MTE	PRN	CSCL	Acc (%)
✓	–	–	–	87.8
✓	✓	–	–	88.1 (↑ 0.3)
✓	✓	–	✓	88.3 (↑ 0.5)
✓	✓	✓	–	88.5 (↑ 0.7)
✓	✓	✓	✓	**89.0 (↑ 1.2)**

Table 4. Comparison of classification accuracies (%) based on (*left*) existing graph contrastive methods and (*right*) different enhanced topologies for the network.

Methods	Acc (%)
Baseline	87.8
w/ SkeletonGCL [8]	88.0
w/ FR-Head [25]	88.2
Ours	**89.0**

Methods	Acc (%)
ProtoGCN	**89.0**
w/o $\mathbf{A}_{\mathbf{intra}}$	88.8
w/o $\mathbf{A}_{\mathbf{inter}}$	88.5
w/o $\mathbf{A}_{\mathbf{intra}}, \mathbf{A}_{\mathbf{inter}}$	88.4

Table 5. Average performance (%) on different difficulty level classes sorted by accuracy under the NTU-120 X-Sub setting.

Difficulty Level	Baseline	ProtoGCN	Δ
class 1-10	64.1	67.1	**+3.0**
class 11-30	78.6	81.1	**+2.5**
class 31-60	88.6	89.7	**+1.1**
class 61-120	96.0	96.4	**+0.4**

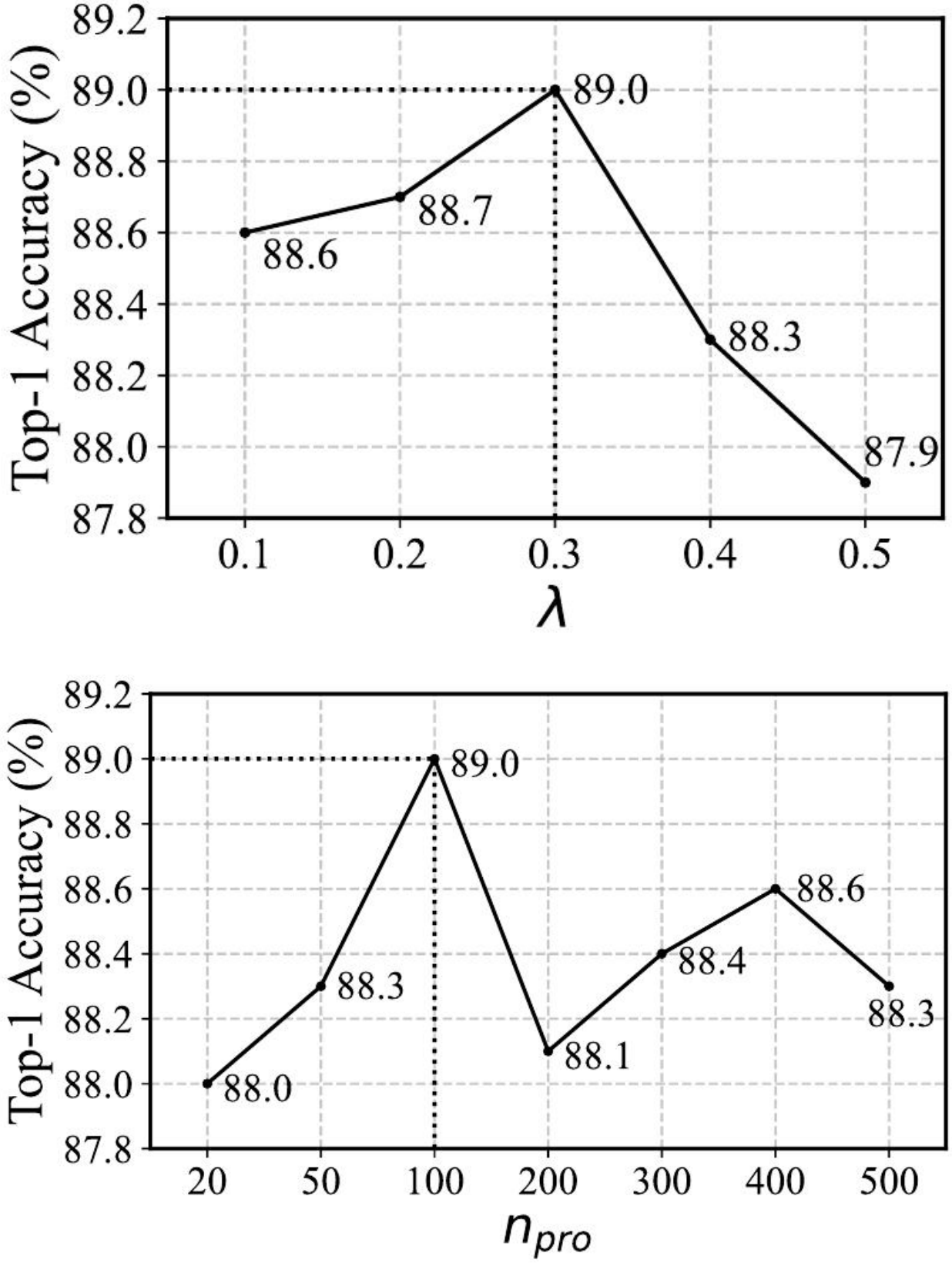

Figure 3. Ablation on the weight λ and memory capacity n_{pro} under NTU-120 (X-Sub).

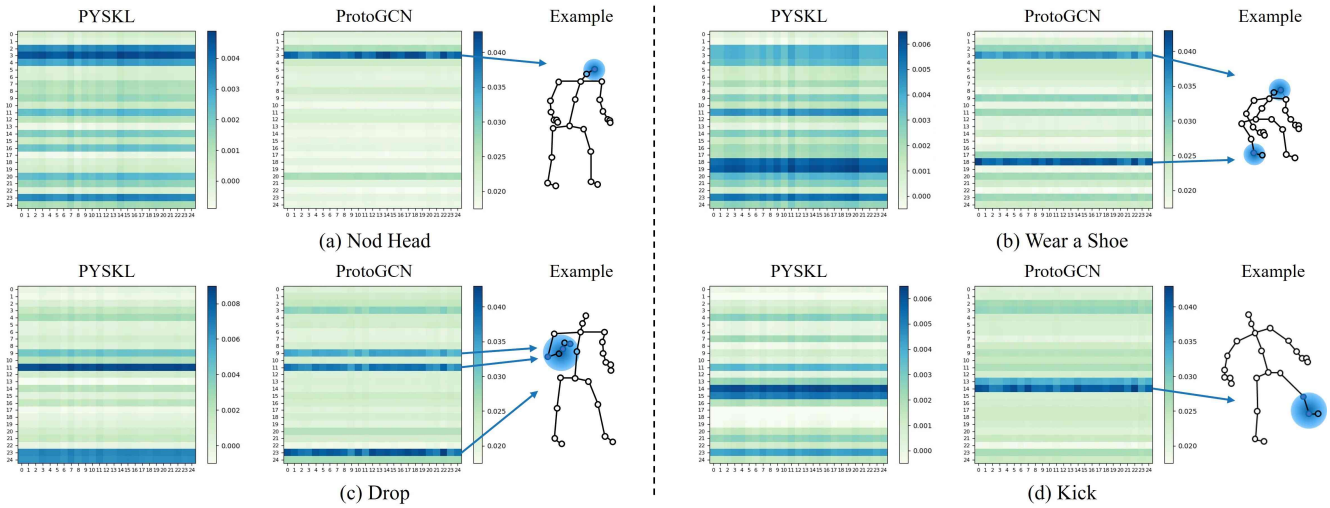

Figure 4. Topologies learned by PYSKL [6] and ProtoGCN across four actions.

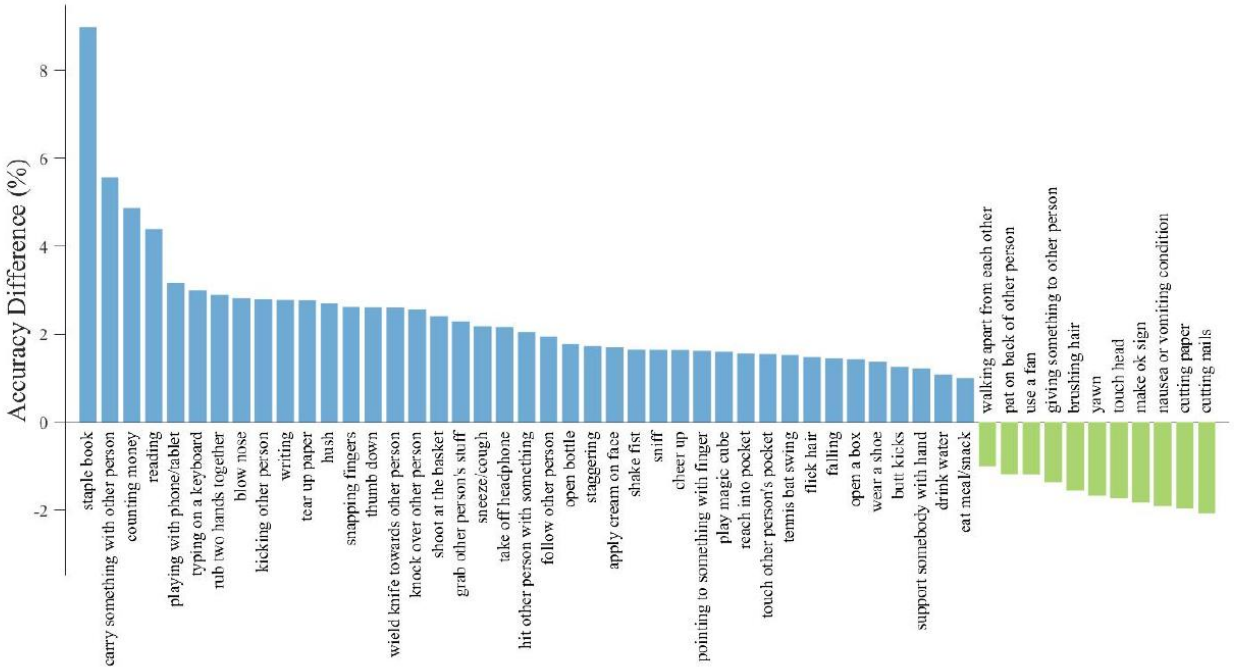

Figure 5. Action classes with accuracy difference greater than 1% between ProtoGCN and PYSKL [6] on NTU-120.